GNN Opening Analysis: Where Do Humans Leave Value on the Table?

Background

In Catan, each player places two settlements before the game starts. **v1** is the first settlement; **v2** is the second. Seat order follows a snake draft: seats 0-1-2-3 place v1 in order, then 3-2-1-0 place v2. Seat 0 picks first but places v2 last; seat 3 picks back-to-back in the middle.

We ask: given where a player placed v1, how suboptimal was their v2 choice?

Model

We trained a **StaticBoardGCN** (a graph convolutional network) on 135,835 labelled openings from 43,947 real games. Unlike prior models that use hand-crafted features, the GNN sees the raw board graph: which vertices neighbour which hexes.

Model	Test AUC
Pip count heuristic	0.503
Logistic regression	0.508
Gradient boosting	0.510
StaticBoardGCN (GNN)	**0.510**

All models are close to 0.5. Win probability from a single opening in a 4-player game is genuinely hard to predict. The GNN's incremental gain confirms that spatial board structure carries a small but real signal.

Method

For each of the **16,978 test-set openings**:

1. Fix the human's v1 placement.
2. Score all 54 possible v2 placements with the GNN.
3. Compute regret = best available score - actual score.

This gives **GNN-conditional v2 regret**: how much win probability did the human sacrifice in their second placement?

Findings

1. Humans leave ~0.3pp on the table in the second placement

Metric	Value
Mean GNN v2 regret	**0.0031**
25th / 75th percentile	0.0019 / 0.0042
% choosing optimal v2	**10.8%**

The distribution is right-skewed: most players are close to optimal, but roughly 15% make choices the GNN considers substantially worse.

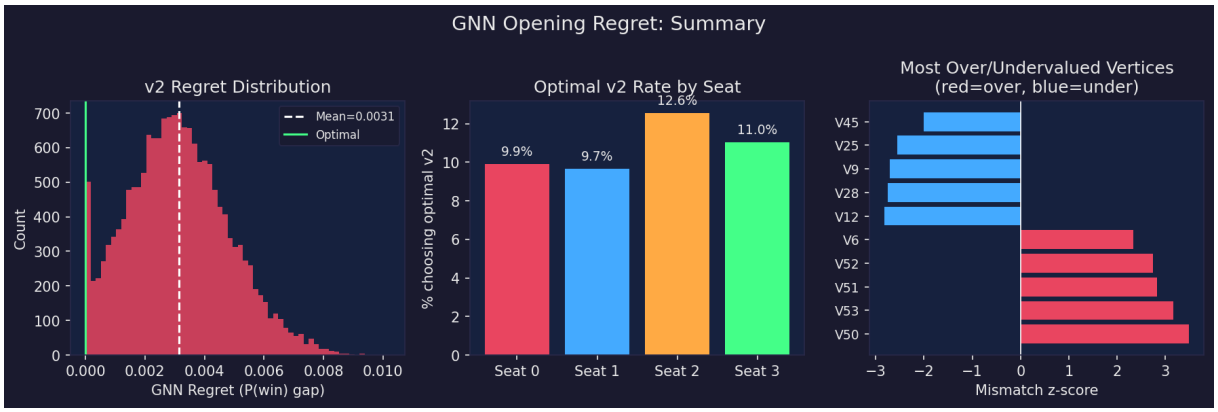

Figure 1: Regret distribution (left), optimal v2 rate by seat (centre), most over/undervalued vertices (right).

2. Human placement frequency shows clear hot spots

Interior vertices (adjacent to 3 hexes, higher pip counts) attract far more v2 placements than rim vertices, regardless of the specific board.

3. The GNN's vertex preferences are narrow but consistent

The GNN's average win probability by v2 vertex spans only **0.2662-0.2674** (~0.12pp). This reflects Catan's balance: no single vertex is dramatically dominant. But the spread is real and consistent with the non-random AUC.

4. Regret is spatially structured

Dot size represents how often humans choose each vertex; colour represents average regret. High-frequency rim vertices tend to carry higher regret: humans choose them often, but the GNN rates them relatively poorly compared to interior vertices.

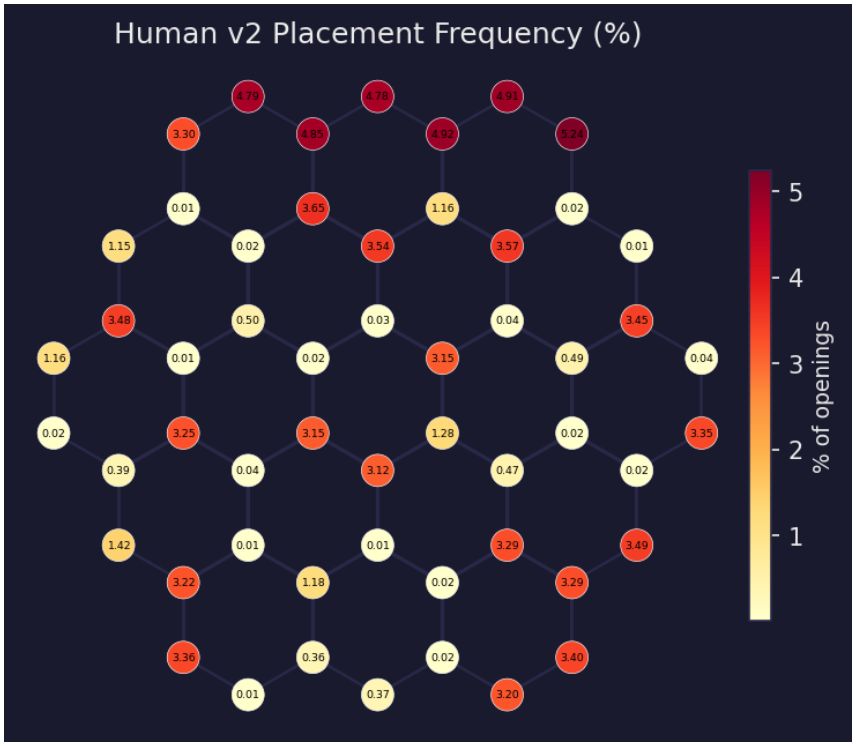

Figure 2: Human v2 placement frequency across all test-set openings.

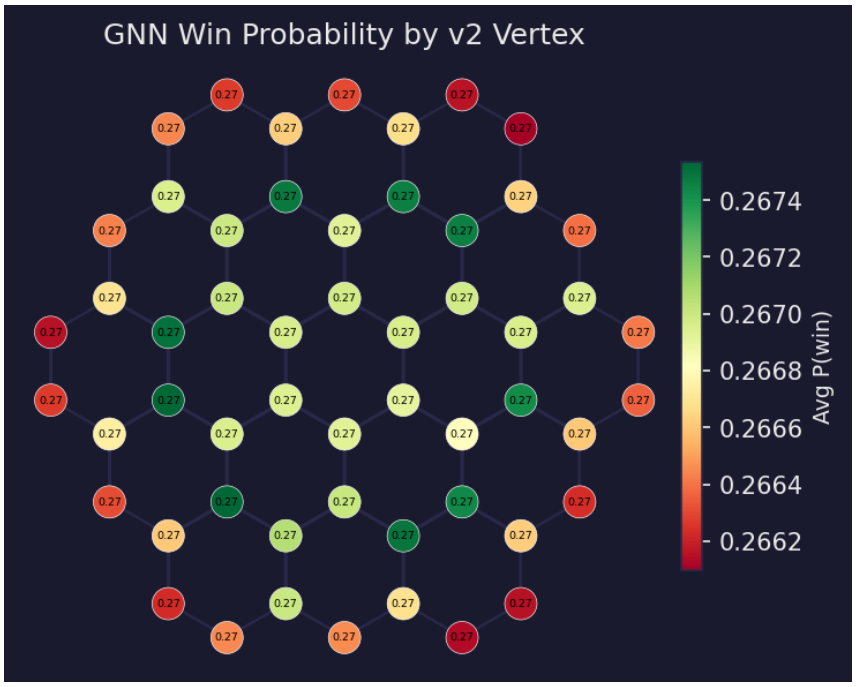

Figure 3: Average GNN-predicted win probability for each v2 vertex.

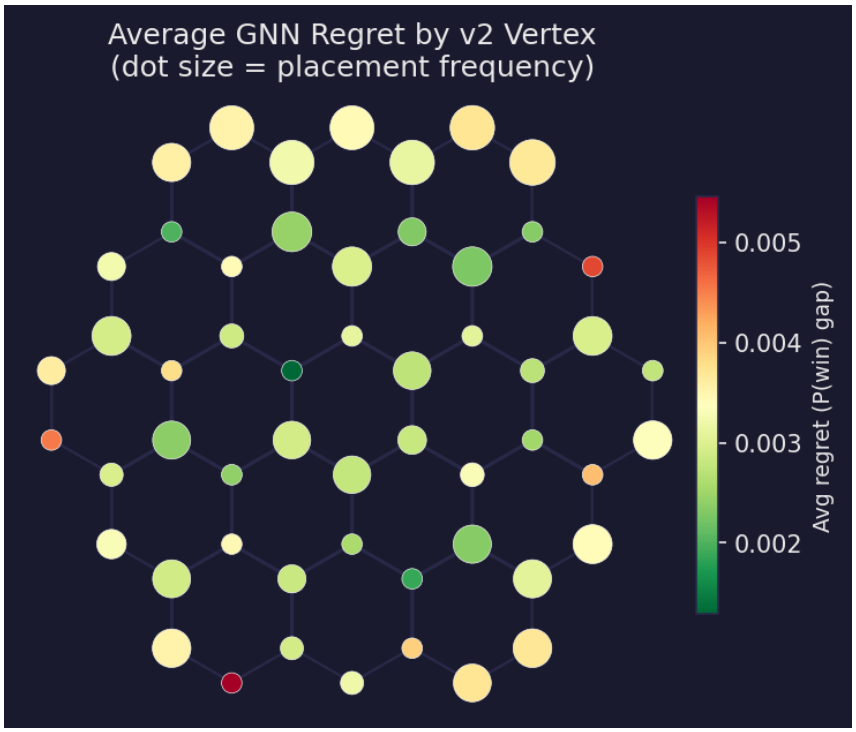

Figure 4: Average GNN regret by v2 vertex. Dot size = placement frequency.

5. Human-GNN mismatch: overvalued ports, undervalued interior

The three panels below show human frequency, GNN value, and the standardised mismatch (positive = humans overvalue, negative = humans undervalue).

- **Overvalued by humans:** port-adjacent rim vertices. Visually salient due to port icons, but the GNN discounts them in v2 because they narrow your production profile.
- **Undervalued by humans:** interior vertices adjacent to 3 hexes. Less obvious visually, but higher expected pip count and better expansion potential.

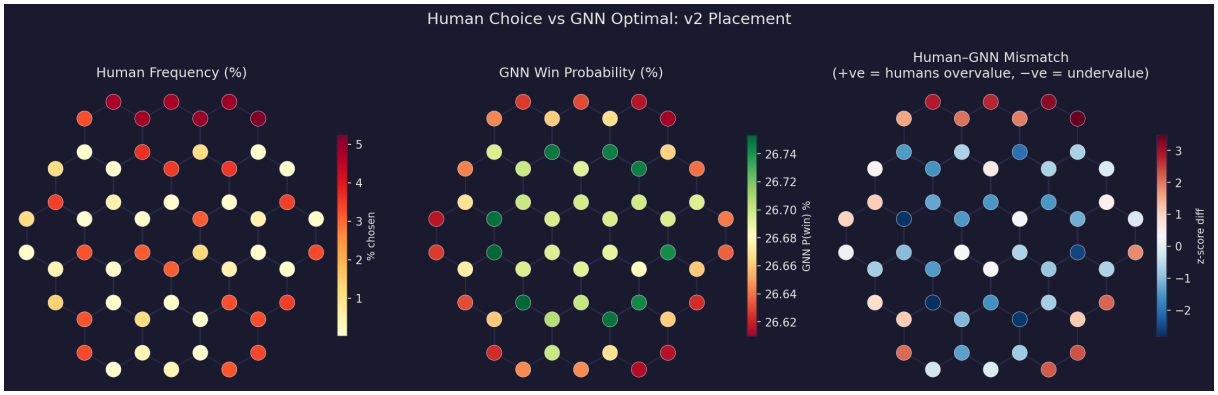

Figure 5: Human frequency (left), GNN win probability (centre), mismatch z-score (right).

6. Seat effects are modest

Seat	% choosing optimal v2
0	9.9%
1	9.7%
2	11.0%
3	12.6%

Later seats choose the GNN-optimal v2 slightly more often, likely because a constrained choice space incidentally aligns with GNN rankings.

Interpretation

Human v2 choices diverge from the GNN-optimal in a spatially consistent way: port-adjacent and visually prominent rim spots are systematically overvalued; interior high-production vertices are undervalued.

The magnitude is modest (0.3pp mean regret) and the opening is not the primary driver of Catan outcomes. But the pattern is meaningful: it reflects a bias toward salient features (ports, isolated spots) over raw production and expansion potential.

The GNN's 0.510 AUC confirms that even with full board topology, the signal from a single opening is weak. The value of this model is less in prediction and more in diagnosis: it provides a topology-aware baseline for what an optimal player would do, against which human biases can be measured.